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Role of radical awareness in the character and word acquisition of Chinese children

A fundamental feature of languages is that groups of words share morphological features. For example, in alphabetic languages like English, words such as *worker, worked*, and *workshop* share the stem *work*, and their meanings are related. In nonalphabetic languages like Chinese, the words 电视 (television), 电话 (telephone), and 电影 (movie) share the character 电 (electric), and they are semantically related.

Morphological relationships among words influence the way words are represented in memory and the process by which skilled readers recognize complex words and derive their meanings (Anshen & Aronoff, 1988; Nagy, Anderson, Schommer, Scott, & Stallman, 1989; Taft, 1985). This conclusion has been reached in studies involving several languages (Feldman & Fowler, 1987; Grainger, Cole, & Segui, 1991; Schriefers, Friederici, & Graetz, 1992). There is also evidence that morphology may influence children's vocabulary acquisition (Tyler & Nagy, 1989; White, Power, & White, 1989; Wysocki & Jenkins, 1987); however, this research has been done only with English-speaking children. The present study seeks to answer this question: Does the morphological structure of Chinese words influence Chinese children's character and word acquisition? If it does, in what ways?

English-speaking children begin to acquire some knowledge of morphology before entering school (Berko, 1958), and this knowledge continues to develop during the school years (Freyd & Baron, 1982; Tyler & Nagy, 1989; Wysocki & Jenkins, 1987). In a pioneering study, Freyd and Baron (1982) investigated whether above-average American fifth graders are more likely than average eighth graders to figure out word meanings through analyzing words into roots and suffixes. They asked students to supply definitions for a list of morphologically simple words (e.g., *vague*) and for a list of derived words (e.g., *acceptable*). Then, using a paired-associate learning task, they asked students to learn and recall pairs of nonsense words. Half of the pairs were related by consistent derivational rules (e.g., *skaf* = steal, *skaffist* = thief), and half were unrelated (e.g., *jeve* = study, *kruttist* = pupil). Good and average students performed equivalently on the morphologically simple words. However, good students did better on the derived words.

In a more recent investigation, Wysocki and Jenkins (1987) taught fourth-, sixth-, and eighth-grade American students the meanings of infrequent words such as *stipulate*, then tested their knowledge of derivatives such as *stipulation*. Students were able to use morphological information to recognize the relationship between the taught words and their derivatives. Sixth and eighth graders were more skilled than fourth graders in using morphological clues.

One of the most sophisticated studies of the acquisition of English morphology was completed by Tyler and Nagy (1989), who tested fourth-, sixth-, and eighth-grade

Role of radical awareness in the character and word acquisition of Chinese children

A TOTAL of 292 Chinese children in the first, third, or fifth grade participated in one of two experiments investigating *radical awareness;* that is, the insight that a component of Chinese characters, called the radical, usually provides information about the character's meaning. The technique used in the experiments was to present two-character words familiar from oral language but which the children had not seen before in print. One of the characters was written in Pinyin, the alphabetic system that Chinese children learn in the first 2 months of first grade. The children's task was to select a character to replace the Pinyin. The first experiment showed that third graders and fifth graders were able to select characters containing the correct radicals even when the characters as a whole were unfamiliar to them, which must mean that they were aware of the relationship between a radical and the meaning of a character. The second experiment showed that children were better able to use radicals to derive the meanings of new characters when the radicals were familiar and the conceptual difficulty of the words was low. Children rated as good readers by their teachers displayed more awareness of radicals than children rated as poor readers.

El rol de la conciencia de los radicales en la adquisición de caracteres y palabras en niños chinos

UN TOTAL de 292 ni os chinos de primero, tercero o quinto grado participó en uno de dos experimentos que investigaron la *conciencia de los radicales*; es decir la comprensión de que un componente de los caracteres chinos, llamado el radical, generalmente proporciona información acerca del significado del carácter. La técnica utilizada en los experimentos consistió en presentar palabras de dos caracteres, familiares en la lengua oral, pero que los ni os no habían visto escritas. Uno de los caracteres fue escrito en Pinyin, el sistema alfabético que los ni os chinos aprenden en los primeros dos meses de primer grado. La tarea de los ni os fue seleccionar un carácter para reemplazar el Pinyin. El primer experimento mostró que los ni os de tercero y quinto grado fueron capaces de seleccionar caracteres que contenían los radicales correctos, aún cuando los caracteres como totalidad no les eran familiares, lo que sugiere que eran conscientes de la relación entre un radical y el significado de un carácter. El segundo experimento mostró que los ni os se desempe aron mejor utilizando los radicales para derivar los significados de caracteres nuevos cuando los radicales eran familiares y la dificultad conceptual de las palabras era baja. Los ni os considerados buenos lectores por los docentes demostraron más conciencia de los radicales que los ni os considerados malos lectores.

Die Bedeutung von etymologischem Bewußtsein beim Erwerb der Sprach- und Wortzeichen bei chinesischen Kindern

EINE GRUPPE von insgesamt 292 Kindern der ersten, dritten oder fünften Schulstufe nahm an einem von zwei Experimenten zur Erforschung des etymologischen Bewußtseins teil; darunter ist die Einsicht zu verstehen, daß ein Bestandteil eines chinesischen Schriftzeichens, nämlich das sinnbildliche Wurzelelement, Radikal genannt, gewöhnlicherweise Informationen zur Bedeutung des Wortzeichens enthält. Die Vorgangsweise, die bei den Experimenten angewendet wurde, war, Wörter mit zwei Schriftzeichen zu präsentieren, die den Kindern vom mündlichen Sprachgebrauch her vertraut waren, aber deren Druckbild sie zuvor noch nicht gesehen hatten. Eines der Schriftzeichen war in Pinyin geschrieben, dem alphabetischen System, das Kinder in den ersten zwei Monaten auf der fünften Schulstufe lernen. Die Aufgabe der Kinder war es, ein Bildzeichen herauszufinden, das das Pinyinzeichen ersetzt. Das erste Experiment zeigte, daß die Kinder der dritten und fünften Schulstufe imstande waren, jene Schriftzeichen auszuwählen, die das richtige Radikal enthielten, auch wenn das Schriftzeichen als Ganzes ihnen unbekannt war, was die Schlußfolgerung nahelegt, daß ihnen die Beziehung zwischen einem Radikal und der Bedeutung eines Zeichens bewußt ist. Das zweite Experiment zeigte, daß die Kinder besser imstande waren, die Bedeutungen von neuen Schriftzeichen abzuleiten, wenn ihnen die Radikale vertraut waren und der Schwierigkeitsgrad in der Sinnerfassung niedrig war. Jene Kinder, die von ihren Lehrern als gute Leser eingestuft wurden, zeigten mehr Bewußtsein für die sinnbildlichen Wurzelelemente als sogenannte schlechte Leser.

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部首を意識することが中国人児童の文字や単語の習得に果たす役割

部首に対する意識調査を目的とした２つの実験のうちの１つに小学１年、３年、５年の計２９２名の中国人児童が参加した。その意識とは部首と呼ばれる漢字の構成要素が多くの場合その文字の意味に関する情報を与えるという洞察のことである。この実験で、児童たちは会話ではよく耳にするが、活字では見たことのない２文字の単語が与えられた。２文字のうちの１文字は、中国人児童が小学校入学時の最初の２ヶ月間で習うピンイン式と呼ばれるローマ字表記法で書かれており、児童に与えられた課題はそのローマ字表記に該当する漢字を選ぶことであった。最初の実験では、漢字があまり馴染みのないものであっても、３年生と５年生は妥当な部首を持つ漢字を選ぶことができた。このことはその児童たちが部首と漢字の意味との関係を認識していることを示すものである。また部首が馴染み深く、単語の概念的難易度が低い場合には部首を使って新しい漢字の意味をうまく引き出すことができるということが２番目の実験で分かったことである。さらに教師によって読みが上手と評価されている児童の方がそうでない児童に比べ、部首に対する意識が高いことも明らかになった。

Rôle de la conscience du radical dans les caractères et l'acquisition des mots chez des enfants chinois

UN TOTAL de 292 enfants de 1°, 3°, et 5° année ont participé à l'une ou l'autre de deux expériences portant sur la *conscience du radical*, à savoir l'intuition qu'une composante des caractères chinois, que l'on appelle le radical, apporte généralement une information sur la signification du caractère. La technique utilisée dans ces expériences a consisté à présenter des mots de deux caractères, connus à l'oral, mais que les enfants n'avaient jamais vus à l'écrit. Un des deux caractères était écrit en pinyin, le système alphabétique que les enfants chinois apprennent dans les deux premiers mois de la 1° année. La tâche des enfants consistait à sélectionner un caractère substituable au pinyin. La première expérience a montré que les enfants de 3° et 5° année étaient en mesure de sélectionner des caractères contenant les radicaux corrects, même quand les caractères dans leur ensemble ne leur étaient pas familiers, ce qui signifierait qu'ils étaient conscients de la relation entre un radical et la signification d'un caractère. La seconde expérience a montré que les enfants étaient plus en mesure d'utiliser des radicaux pour en déduire les significations des nouveaux caractères quand les radicaux étaient familiers et que la difficulté conceptuelle des mots était faible. Les enfants considérés comme de bons lecteurs par leurs maîtres ont manifesté une plus grande conscience des radicaux que les enfants considérés comme de mauvais lecteurs.

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American students for knowledge of different aspects of derivational suffixes. Knowledge of morphology increased with grade. When given a low-frequency derivative of a high-frequency word, fourth graders could recognize the relationship between unfamiliar derivatives and known words. However, knowledge of the syntactic function of derivational suffixes and knowledge of distributional constraints on their use increased through the eighth grade. Like Freyd and Baron (1982), Tyler and Nagy (1989) found that students of above-average ability made more use of morphology than students of lower ability.

Extending the research of Tyler and Nagy (1989), Nagy, Diakidoy, and Anderson (1993) explored the development of knowledge of the meanings of 10 common English suffixes among 630 fourth-grade, seventh-grade, and high school students. They found that knowledge of derivational suffixes underwent significant development between fourth grade and high school. Even in high school, however, there were some students who showed little knowledge of suffixes. While knowledge of suffixes was related to general verbal ability, it seemed to be a distinct component of skilled reading.

Many words in English, particularly infrequent words, can be divided into roots and affixes. The meaning of a derived or compound word is not always predictable from the meanings of its components. Nonetheless, a great many English words can be figured out at least in part from their roots, prefixes, and suffixes (Nagy & Anderson, 1984). For this reason, morphological analysis is generally assumed to be one of the cornerstones of vocabulary development among English-speaking children (Nagy & Anderson, 1984; White et al., 1989; Wysocki & Jenkins, 1987). Chinese linguists and educators also assume that knowledge of morphology contributes to character and word acquisition in Chinese, although there is little empirical research to support this assumption (Hoosain, 1991).

Chinese is a logographic writing system in which the written symbols, or characters, represent lexical morphemes. Chinese words have two levels of morphological structure. First, the majority of Chinese words are compounds made up of two or more characters, each of which represents an independent meaning. For example, in the two-character word 牛奶 (cattle-milk), the meaning of the whole word can be readily understood by combining the meanings of the two separate characters. In Chinese, words that share the same character usually are semantically related, for example, 牛奶 (cattle-milk), 牛肉 (cattle-meat), 牛油 (cattle-oil). Of course, the relationship between the meaning of a word and its component characters is not always this regular.

Another level of analysis of Chinese involves the internal structure of characters. About 80%–90% of the characters in modern Chinese are composed of two components: a component called a *radical* that gives a clue to meaning and a component that offers a clue to pronunciation (Hoosain, 1991). The radical might itself be an independent character or it might be a bound form that occurs only within characters. For example, the character 松 (pine) contains the 木 (wood) radical, which is itself a character, whereas the character 河 (river) contains the 氵 (water) radical, which is not a free-standing character.

Chinese has more homophones than most languages. Compared to the tens of thousands of potential English syllables, Chinese has only about 400 possible syllables (or about 1,200 when tones are considered). The great numbers of homophones would cause difficulties in reading comprehension if there were no radicals to provide a way to separate and disambiguate homophone characters and words. So, the internal structure of characters certainly plays an important role in skilled Chinese reading.

Large groups of characters, sometimes numbering more than 100, share the same radical. For example, the characters 吠 (bark), 吻 (kiss), 喊 (shout), 唱 (sing), and 喝 (drink) share the radical 口 (mouth), and each is clearly related in meaning to mouth. In most cases, the radical offers some basis for inferring the meaning of a character. However, a radical does not always provide a reliable guide to meaning. Over centuries of use, the Chinese writing system has changed, and the inevitable result is that there are irregular characters whose meanings are unrelated to the meanings of their radicals, or only indirectly related. In regular cases, the meaning relation between the radical and the character can be called *transparent* while in irregular cases it can be called *opaque* (Flores d'Arcais, 1992; T'sou, 1981).

Both levels of morphological structure of Chinese have been found to affect the processing of words by skilled readers. In several recent studies, the reaction time for recognizing compound words was influenced by the frequency of the component characters. For a two-character word, the reaction time was affected more by the frequency of the first character than by the frequency of the second (Taft & Zhu, 1995; Zhang & Peng, 1992; Zhou & Marslen-Wilson, 1994).

Hatano, Kuhara, and Akiyama (1981) asked Japanese students to match compound words, such as *leukemia*, with their definitions. Students performed better when words were presented in Kanji, the Japanese version of characters, for example, 白血病, which literally means white-blood-disease, than in Kana, the Japanese writing system for representing pronunciations. It seems that the morphological information in compound Kanji words helps readers to infer word meanings.

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Zhang, Zhang, and Peng (1990) asked college students to make speeded judgements about whether or not a series of one-character words signified females. The reaction time was shorter when a word had a radical whose meaning was consistent with the meaning of the word than when it had a radical whose meaning was inconsistent with the meaning of the word. One of the former cases is the word 姑 (aunt) which has a 女 (female) radical. The word 婿 (son-in-law) which also has a 女 (female) radical is one of the latter cases.

Miao and Sang (1991) asked students to verify sentences such as "a swallow (target) is a bird (category)." Half of the target words were of high frequency, and half were of low frequency. Three types of sentences were presented. The first type contained target words with radicals consistent with the categories. For example, in the sentence "a lion is an animal," the target 狮 (lion) has an 犭 (animal) radical. The second type contained target words without radicals. For example, in the sentence "a swallow is a bird," the target 燕 (swallow) does not have a 鸟 (bird) radical. The third type contained target words that had radicals inconsistent with the categories. For example, in the sentence "a whale is an animal," the target 鲸 (whale) has a 鱼 (fish) radical. No difference was found in the reaction time among the three types of sentences when the target words were of high frequency. In contrast, when sentences containing low-frequency words were verified, the reaction time for the second and the third type of sentence was much longer than that of the first type.

Whereas most previous empirical studies of Chinese morphology have examined skilled readers, the present study explored the role of morphological knowledge in Chinese children's character and word acquisition. Basic morphological knowledge, only at the level of character radicals, was investigated. That is, the study addressed whether children were able to recognize and make productive use of the relationship between a word and the radical of a character in the word.

A Chinese child has to learn about 3,000 visually complex characters during the first 6 years of school. Children and teachers alike regard this as an extremely difficult task. It might seem that the task could be made less daunting if teachers were to cultivate insights into Chinese morphology. Some, but by no means all, teachers do point out that radicals give clues to meaning; however, we are unaware of any schools that provide systematic instruction in using the information in radicals. As we have already indicated, radicals are an imperfect guide to meaning. Many Chinese teachers are probably suspicious that too much emphasis on using radicals would lead to guessing—and they want students to know the exact meanings of characters, not guess. A comparable case in the United States would be the varying attitudes of teachers toward invented English spellings.

We investigated five questions in two experiments: (a) Are elementary school children aware of the radicals in characters? (b) Does radical awareness help children in remembering familiar characters and learning unfamiliar characters? (c) How do children use the information in radicals to derive the meanings of unfamiliar characters? (d) Is there a developmental progression over the early school years in the use of radicals? and (e) What differences are there between good and poor readers in utilizing radicals to learn characters and words?

Experiment 1

In the first experiment, we concentrated on the basic question: Are children aware of the function of radicals? The technique was to present a series of compound words known to children from oral language in which one of the syllables was represented by a known character and the other syllable by Pinyin. The children's task was to select the appropriate character to replace the Pinyin.

Pinyin is a Chinese alphabetic system that provides the pronunciation of characters. Chinese children learn Pinyin during the first 10 weeks of the first grade before they learn to read characters. Thereafter, they practice Pinyin frequently because Pinyin is used as a guide to pronunciation whenever a new character is introduced in a textbook. Since all of the words used in the experiment were familiar from oral language, we assumed that children would know their meanings when they read them with the assistance of Pinyin.

For each word, four characters with the same sound components but different radicals were presented as options. For example, 瞳 (pupil), 撞 (bump), 僮 (boy servant), and 潼 (place name) were the options to replace the Pinyin in tóng孔 (pupil).

At least two of the characters in each set of four were visually unfamiliar to the children. If children are sensitive to the relationship between a radical and the meaning of a word, they ought to be able to select the correct character, even when it is visually unfamiliar. Thus, the aim of the task was not simply to test knowledge of individual characters or words, but to measure children's radical awareness—their ability to predict the written representation of words they have not previously encountered in print.

Method

Subjects. A total of 220 students, from six classrooms from an elementary school in Beijing, participated

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in this study. Most of the students were from workers' families. Their parents typically were elementary school or middle school graduates. There were 67 first-grade students, 32 boys and 35 girls; 71 third-grade students, 36 boys and 35 girls; and 82 fifth-grade students, 42 boys and 40 girls. For the data analysis, the children were divided into high, average, and low Chinese reading levels on the basis of ratings by their teachers. Formal tests of reading level are rarely used in China, so the teachers' ratings were based on their experience with the children's classroom performance.

At the school in which the study was conducted, class size ranges from 35 to 40 children, who sit in rows of bolted-down double desks. Whole-class, teacher-directed instruction predominates, although there is some individual seat work as well as partner work between children sharing the same desk.

Each day there are two 45-minute Chinese lessons based on the national reading/language arts textbooks used throughout China. Teachers at the school estimate that about 60% of lesson time in the first and second grade is spent introducing, practicing, and reviewing characters. The proportion of time spent directly on characters decreases from grade to grade to about 20% in the fifth grade. At each grade, most of the remaining lesson time is spent in guided reading of stories and other text selections. Teachers at the school reported that they do not provide systematic instruction in radicals or other aspects of Chinese morphology.

Design and procedure. A 3 (grade) × 3 (reading level) × 3 (familiarity of character) × 3 (morphology of character) mixed design was used, in which grade and reading level were between-subject variables and the familiarity and morphology of characters were within-subject variables.

For each grade, there were 90 target characters selected from the national reading/language textbooks. Thirty of the characters were classified as familiar because they had been introduced in the textbook two grades earlier, and the children had encountered them numerous times. For example, the familiar characters for third graders were from the first-grade textbook; for first graders these were characters introduced one semester before. At each grade, 30 characters had been recently learned. These came from the textbook students were using in the present semester. Finally, 30 characters were classified as unfamiliar to students because they would not appear in the textbook until two grade levels above the students' current grade.

For each degree of familiarity, three types of characters were included. Ten characters were morphologically transparent in that their radicals were very helpful in figuring out the meaning of the whole (e.g., the character 烛 [candle] which contains the radical 火 [fire]). Ten were morphologically opaque characters. These contained radicals that contributed little or nothing to word meanings (e.g., the character 错 [mistake] contains the radical 钅 [metal]). The remaining 10 were unanalyzable characters that cannot be broken into components (e.g., the character 欠 [owe] consists of only one part). The morphological transparency of the target characters was rated by three graduate students and researchers on a three-point scale in which 1 stood for opaque and 3 stood for transparent. The characters rated 1 by all three of the raters were classified as morphologically opaque while those rated as 3 by all of them were classified as morphologically transparent.

The target characters were compiled into 90 two-character words familiar from oral language. Each word consisted of one target and one nontarget character. The target appeared in Pinyin and the nontarget as a known character. For each word, children were asked to replace the Pinyin with one of four characters. The distractors were three characters having the same sound components, but different radicals than the correct character. For example, for the item tiào 望 (look into the distance from a high place), the correct answer is the character 眺, which has the radical 目 (eye) and means look from a high place. The three distractors were the character 挑, which has the radical 扌 (hand) and means pick; 跳, which has the radical ⻊ (foot) and means jump; and 佻, which has the radical 亻 (person) and means skittish. For the unanalyzable characters (e.g., 再 [again]), the distractors were three characters that were visually similar to the target (e.g., 冉 [gradually], 甩 [throw off], and 耳[ear]).

The items were presented in the form of a paper-and-pencil multiple-choice test suitable for group administration to an entire classroom of children. An item was scored 1 when the correct answer was selected and scored 0 when a distractor was selected. The data were analyzed using unweighted means analysis of variance. Prior to analysis, the raw proportions were transformed using an arcsin transformation, because some children obtained perfect or near-perfect scores with some types of characters.

Results and discussion

Children's mean proportions correct for the different types of characters are shown in Table 1. It is apparent that children performed better on familiar than less familiar characters. It is also apparent that under most conditions children performed better when the characters were morphologically transparent than when they were either morphologically opaque or unanalyzable.

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Table 1 Mean proportion correct (and SDs) as a function of type of character and grade

		Morphology		
Grade	Familiarity	Opaque	Unanalyzable	Transparent
First	Unfamiliar	.36 (.18)	.57 (.22)	.57 (.21)
	Recently learned	.77 (.16)	.87 (.19)	.84 (.18)
	Familiar	.89 (.16)	.85 (.20)	.92 (.16)
Third	Unfamiliar	.37 (.16)	.55 (.22)	.77 (.19)
	Recently learned	.46 (.20)	.61 (.21)	.76 (.20)
	Familiar	.83 (.14)	.94 (.12)	.96 (.08)
Fifth	Unfamiliar	.42 (.19)	.57 (.15)	.70 (.11)
	Recently learned	.72 (.18)	.84 (.14)	.97 (.07)
	Familiar	.94 (.09)	.96 (.07)	.98 (.05)

Analysis of variance confirmed that the main effect of character familiarity was statistically significant, $F(2, 416) = 1490.81$, $p < .01$, $MSE = .08$. That is, children obtained high scores on familiar characters, but their performance declined as the familiarity of characters decreased.

The main effect of morphology of characters was statistically significant, $F(2, 416) = 513.07$, $p < .01$, $MSE = .06$. The mean proportion correct on morphologically transparent characters (.83) was much higher than that on the unanalyzable (.75) or opaque characters (.64). These results imply that children are aware of radicals and that radical analysis is helpful in their character learning.

The interaction of morphology and familiarity, $F(4, 832) = 53.63$, $p < .01$, $MSE = .06$, was statistically significant. Table 2 suggests that radical analysis contributed differently depending upon character familiarity. When the characters were familiar to children, the difference between morphologically transparent characters and the other two types of characters was relatively small, although it was statistically significant, $F(1, 208) = 42.96$, $p < .01$, $MSE = .03$. Children performed much better on morphologically transparent characters than on unanalyzable and opaque characters, both when the characters were recently learned, $F(1, 208) = 281.72$, $p < .01$, $MSE = .06$, and especially when the characters were unfamiliar to them, $F(1, 208) = 402.86$, $p < .01$, $MSE = .07$. Similar interactions between morphological structure and frequency of exposure to Chinese words or characters have been reported in experiments with adults (Miao & Sang, 1991; Seidenberg, 1985).

The fact that the importance of morphology became larger as character familiarity decreased may mean that children do not routinely decompose familiar characters into components in order to access the internal lexicon. Or, it may mean that they have encoded elements of familiar characters that are not systematically related to morphology. But the performance of children on less familiar characters clearly shows an influence from morphological structure. The children must have identified radicals and recognized the relationship between the meanings of radicals and the meanings of words containing the radicals. The results imply that children use the information in radicals to learn and remember characters recently introduced in school and to make inferences about the meanings of unfamiliar characters.

A statistically significant interaction of grade by morphology by familiarity, $F(8, 832) = 9.11$, $p < .01$, $MSE = .06$, suggests a developmental progression in the use of information in radicals. First graders did not yet show a clear ability to utilize the semantic information in radicals. In contrast, third and fifth graders assuredly were using radical analysis to derive unfamiliar characters as well as to learn and remember characters recently introduced in school.

There was a statistically significant interaction of familiarity and reading level, $F(4, 416) =14.43$, $p < .01$, $MSE = .08$. Table 3 indicates that children with different reading levels did equally well on the familiar characters,

$F(2, 208) = 2.55$, $p > .05$, $MSE = .08$, but as familiarity decreased, the performance of children with higher and lower reading levels diverged. Better readers got statistically significantly higher scores than did poorer readers on recently learned characters, $F(2, 208) = 35.97$, $p < .01$, $MSE = .24$, and also on unfamiliar characters, $F(2, 208) = 25.44$, $p < .01$, $MSE = .24$. Two explanations are possible. First, better readers may have more specific character knowledge than poorer readers, including knowledge of some characters not yet introduced in school. Alternatively, better readers may have more facility than poorer readers have in decomposing complex characters and using morphological knowledge to assimilate new characters.

The ANOVAs reported up to this point were conventional ones in which the subject was the unit of analysis. Parallel ANOVAs were conducted using the word as the unit of analysis—that is, the data were aggregated by the word rather than by subject. The results were almost identical to the ones in which the subject was the unit of analysis. The main effects of character familiarity, $F(2, 783) = 246.57$, $p < .01$, $MSE = .21$, and the main effect of character morphology, $F(2, 783) = 61.41$, $p < .01$, $MSE = .21$, were both statistically significant. The interaction of morphology and familiarity was also statistically significant, $F(4, 783) = 6.02$, $p < .01$, $MSE = .21$, as was the interaction between familiarity and reading level, $F(4, 783) = 3.22$, $p = -.02$, $MSE = .21$. The similarity of the analyses by subject and by word suggests that the findings are robust and generalizable.

About half of the target characters (.48) contained a radical, such as 火 (fire), which can itself be a character, while the other half of the characters contained radicals, such as 扌 (hand), which are bound forms that do not themselves constitute characters. One might have supposed that it would be easier for children to infer the meanings of radicals that can be characters, but the results showed no difference between the two types of radicals, $F(1, 534) = .37$, $p > .05$, $MSE = .03$, or any interaction between type of radical and the children's rated reading level, $F(2, 534) = .04$, $p > .05$, $MSE = .01$.

Table 2 Mean proportion correct (and SDs) as a function of character familiarity and morphological structure

	Morphology		
Familiarity	Opaque	Unanalyzable	Transparent
Unfamiliar	.38 (.18)	.56 (.19)	.68 (.19)
Recently learned	.65 (.22)	.77 (.21)	.86 (.18)
Familiar	.89 (.14)	.92 (.14)	.95 (.10)

Table 3 Mean proportion correct (and SDs) as a function of character familiarity and reading level

	Reading level		
Familiarity	Low	Average	High
Unfamiliar	.45 (.17)	.55 (.17)	.63 (.19)
Recently learned	.64 (.23)	.79 (.23)	.83 (.18)
Familiar	.89 (.15)	.93 (.13)	.94 (.12)

Experiment 2

In Experiment 2, the process of morphological analysis was explored further. Nagy and Scott (1990) proposed that general knowledge of word structure, knowledge of specific roots, prefixes, and suffixes, and strategies for using the knowledge when encountering new words are important in morphological analysis of English words. Similarly, we assume that analyzing an unfamiliar character into its parts, accessing the semantic information in the radical, and using the information in the radical to figure out the meaning of the word are required for productive morphological analysis of Chinese words. Specifically, in Experiment 2, we aimed to get partial answers to two questions: (a) How do children use radicals to derive the meanings of unfamiliar characters? and (b) What are the differences between good and poor readers in radical analysis?

Because it has been found that morphological analysis is more likely, and more important, when processing low-frequency words (Miao & Sang, 1991; Seidenberg, 1985; Shu & Zhang, 1987), in this experiment all of the targets were unfamiliar characters. We varied whether the target characters had familiar or unfamiliar radicals. For example, the radical 隹 of the character 雏 (young bird) means bird, but few people know it. We also varied whether the words containing the target characters were associated with easy or with difficult concepts. We hypothesized that if the familiarity of the radical and the conceptual difficulty of word are important in the process of figuring out unfamiliar characters, performance will be heavily affected when either of the factors is changed.

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Table 4 Mean proportion correct (and SDs) as a function of morphology and conceptual difficulty

	Morphology		
Conceptual difficulty	Opaque	Transparent Unfamiliar radical	Transparent Familiar radical
Difficult	.28 (.13)	.30 (.17)	.39 (.15)
Easy	.35 (.17)	.40 (.19)	.62 (.21)

Method

Subjects. The subjects were 39 third- and 33 fifth-grade students from the same school as the subjects in Experiment 1. The children were rated by their teachers as high, average, or low in reading level.

Design and procedure. A 3 (reading level) × 2 (conceptual difficulty of word) × 3 (morphology of character) mixed design was used. Reading level was a between-subject variable, and the conceptual difficulty of words and the morphology of characters were within-subject variables.

The format of the multiple-choice test students received and the procedures for measurement were similar to that used in the first experiment. All of the target characters in the multiple-choice test were unfamiliar to children. The target characters for third graders were selected from the Chinese reading/language textbooks for fourth, fifth, and sixth grades, and the targets for fifth graders were from the textbooks for seventh, eighth, and ninth grades.

For each grade, there were 60 target characters of three types. Twenty characters were morphologically transparent and contained radicals familiar to children (e.g., the character 聆 [listen respectfully] has the radical 耳 [ear]). Twenty were morphologically transparent but contained radicals unfamiliar to children (e.g., the character 雏 [young bird] has the radical 隹 [bird]). The remaining 20 characters were morphologically opaque (e.g., the character 辑 [edit] has the radical 车 [vehicle]). Within each of these three types of characters, 10 were formed into conceptually easy words and 10 into conceptually difficult words.

The familiarity of radicals was rated on a two-point scale by two school teachers. The radicals rated as 1 by both of the teachers were included in characters with unfamiliar radicals, while radicals rated as 2 by both teachers were included in characters with familiar radicals. Similarly, the words formed using the target characters were rated on another two-point scale, in which two school teachers were asked to check whether a word was in children's oral vocabulary or, if not, whether the meaning of the word would be readily understandable to children because it involved familiar concepts. The words rated 1 by both of the teachers were classified as conceptually difficult, while the words rated 2 by both of the teachers were classified as conceptually easy. As in Experiment 1, the data were analyzed in an unweighted means analysis of variance following an arcsin transformation. In the primary analysis, the unit of analysis was the individual subject's performance.

Results and discussion

The main effect of the conceptual difficulty of the words was statistically significant, F (1, 64) = 85.07, $p < .01$, $MSE = .03$. Children's performance was better on characters that were constituents of easy words (.46) than on ones that were constituents of difficult words (.32). This result is consistent with the previous studies with English-speaking children (Nagy, Anderson, & Herman, 1987).

The main effect of morphology was also statistically significant, F (2, 128) = 37.94, $p < .01$, $MSE = .02$. Children's performance on transparent characters with familiar radicals was considerably higher (.50) than on transparent characters with unfamiliar radicals (.35) or opaque characters (.32). This confirms the finding of Experiment 1 that third and fifth graders indeed use the information in radicals when they encounter new characters.

The set of transparent characters with unfamiliar radicals provides an additional control beyond those employed in Experiment 1. The relatively poor performance of the children on this set strengthens the conclusion that it is the analysis of radicals that is responsible for the good performance on transparent characters with familiar radicals.

An interesting interaction appeared between conceptual difficulty and morphology, F (2, 128) = 7.07, $p < .01$, $MSE = .01$. Table 4 displays the interaction. What stands out is the children's superior performance on the conceptually easy words when the characters were morphologically transparent and had familiar radicals. Neither a conceptually easy word nor accessible mor-

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Table 5 Mean proportion correct (and SDs) as a function of morphology and reading level

	Morphology		
Reading level	Opaque	Transparent Unfamiliar radical	Transparent Familiar radical
Low	.36 (.16)	.34 (.19)	.39 (.16)
Average	.30 (.15)	.33 (.16)	.53 (.17)
High	.33 (.13)	.41 (.20)	.60 (.17)

phology was very helpful alone. Evidently the process of deriving the meaning of an unfamiliar character is strongly facilitated when a child is able to use both sources of information.

A statistically significant interaction of reading level with morphology was obtained in this experiment, $F(4, 128) = 4.22$, $p < .01$, $MSE = .02$. Table 5 shows that there is no difference in performance on opaque characters among children with different reading levels, $F(2, 64) = 2.68$, $p > .05$, $MSE = .02$. Similarly, there was no statistically significant difference on morphologically transparent characters with unfamiliar radicals as a function of reading level. However, on morphologically transparent characters with familiar radicals, high, average, and low readers performed quite differently, $F(2, 64) = 9.65$, $p < .01$, $MSE = .04$.

If the advantage of better readers was attributable to their greater knowledge of specific characters, then they would have done better than poorer readers on all three types of characters. In fact, they scored higher only on characters with accessible morphology. Therefore, it appears that the explanation for the advantage is that average and, especially, good readers make use of morphology, while poor readers cannot or do not.

Parallel analyses using the word as the unit of analysis yielded results almost identical to the ones in which the subject was the unit of analysis. The main effect of the conceptual difficulty of words, $F(1, 162) = 26.26$, $p < .01$, $MSE = .14$, was statistically significant, as was the main effect of morphology, $F(2, 162) = 22.24$, $p < .01$, $MSE = .14$. Statistically significant interactions were found between conceptual difficulty and morphology, $F(2, 162) = 3.33$, $p = .038$, $MSE = .14$, and between morphology and reading level, $F(4, 162) = 3.04$, $p = .019$, $MSE = .14$. Again, the converging evidence from the two types of analysis strengthens confidence in the findings.

General discussion

The major finding of these experiments is that many Chinese children have a functional awareness of the relationship between the radical in a character and the meaning of words containing the character. Knowledge of morphology was found to develop with grade level. It was not clearly found among first graders but was plainly evident among third graders and fifth graders. Probably children's knowledge of morphology increases as they have more experience with printed language.

First graders may treat characters as unanalyzed wholes, or at least they do not make systematic use of the components of characters used by skilled readers of Chinese. This would explain why first graders do well on characters familiar to them, but relatively poorly on unfamiliar characters, even ones that are morphologically transparent.

Studies reported in this article clearly show that while third graders and fifth graders might process familiar characters holistically, they are able to decompose characters into informative parts. The key evidence is their superior performance on unfamiliar, morphologically transparent characters, as compared to other kinds of characters for which morphological analysis could not be helpful. Evidently, children as young as third grade are able to differentiate the semantic information provided by radicals, integrate this information with the meanings of words, and successfully infer the meanings of unknown characters.

Experiment 2 established boundary conditions for successful morphological analysis. The results show that knowledge of specific radicals, knowledge of word meanings, and command of the strategy of using morphological analysis are all necessary. First, children must know the meaning of the radical in the new character.

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When the semantic information that a radical provides is vague, or the radical is unfamiliar, children's attempts to derive characters are seriously hindered. Second, the word containing a new character must be conceptually easy. When a new character appears in a conceptually difficult word, regardless of whether the radical provides useful semantic information or whether the radical is familiar, children have a low probability of deriving the character. Third, children must possess the strategy of integrating radicals and word meanings. Even when all of the information for deriving a character was available, poor readers typically did not make the inference. This must mean that making inferences based on morphology is not a strategy that is inevitably acquired as children get older and learn more words.

Importantly, the advantage of the good readers was confined to unfamiliar but transparent characters. Therefore, it appears that good readers can be distinguished from poor readers not just in their vocabulary knowledge, but in their ability to interpret novel characters on the basis of their radicals. Poor readers either have not discovered the basic morphological features of Chinese writing or do not spontaneously use what they know about morphology to derive new characters. A possible practical implication of this finding is that poor readers might benefit from explicit instruction in morphology or strategies for using morphology.

A common assumption among teachers in China and elsewhere is that sustained vocabulary growth depends upon instruction, drill, and practice in school. The present results call this assumption into question. The results suggest that many Chinese children are able to independently learn new characters and words. When they are reading on their own, these children are likely to be able to use their knowledge of morphology and information they glean from the text in order to derive the meanings of new characters and words.

Research establishes that natural learning of word meanings while reading is by far the most important avenue for vocabulary growth among English-speaking children (Anderson & Nagy, 1992). It is a plausible hypothesis that independent reading is a major source of vocabulary growth among Chinese-speaking children, as well.

Indeed, Shu, Anderson, and Zhang (1995) have already reported evidence that favors this hypothesis. They completed a cross-cultural investigation of natural learning of word meanings while reading. A total of 447 Chinese and American children in third and fifth grades read one of two cross-translated stories and then completed a test on the difficult words in both stories. The results showed statistically significant learning of word meanings while reading in both grades in both countries.

In the Chinese part of the Shu et al. study, amount of out-of-school reading was investigated at the beginning of the spring semester. The children were asked to write down the names of books that they had read during the immediately preceding winter vacation. Scores on a vocabulary test were directly proportional to the amount of reading the children reported. Moreover, children who reported reading extensively had a much higher likelihood of learning the meaning of a previously unfamiliar word simply from reading a text containing the word. The likelihood of learning the meaning of a word for children who reported reading eight or more books during winter vacation was over three times as great as for children who read four to seven books and over seven times as great as for children who read three or fewer books. Evidently, Chinese children who read extensively outside of school have *learned how to learn* unfamiliar words. Morphological analysis is implicated as a key element in these word-learning strategies.

Limitations

The present study leaves several questions in need of answers. As we just explained, morphological awareness is strongly implicated as an enabling factor in incidental learning of vocabulary while reading. However, it could be argued that the Pinyin replacement task employed in the present study focused attention on radicals, and that during normal reading children might not pay attention to and use radicals. A direct empirical test is needed of whether radical awareness, as measured by performance on the Pinyin replacement task and other measures, actually predicts incidental learning of vocabulary while reading.

A second question concerns the source of children's awareness of radicals. The study establishes that older children rated as better readers by their teachers have acquired insight into the function of radicals, but leaves open the question of exactly how they acquired this knowledge. In particular, it is difficult to say how much of a role is played by children's self-discovery while reading as compared to classroom instruction in morphology.

A third question left open is whether insights into morphology would be more or less important for children from other parts of China. Children from Beijing, where the study was done, speak standard Chinese, or Mandarin. Regional dialects spoken elsewhere in China are so different from Mandarin that they can be regarded as separate languages. In effect, children outside of Beijing are learning to read in a second language, and one might suppose that for them morphological awareness would be more important but perhaps harder to acquire (see Nagy & Anderson, in press).

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In sum, the present experiments provide evidence that many Chinese children are aware of aspects of the morphological structure of Chinese characters and words. The experiments further show that Chinese children's knowledge of morphology develops over the school years, and there are differences in functional knowledge of morphology associated with reading level. Similar findings have been obtained with English-speaking children (Freyd & Baron, 1982; Nagy et al., 1993; Tyler & Nagy, 1989; Wysocki & Jenkins, 1987). Indeed, despite the vast differences between the writing systems, there seems to be a parallel development in children's knowledge and use of morphology in Chinese and in English.

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AUTHOR NOTE

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